

CA6114 Part 1

Chapter 1 - Introduction

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Chapter 1 – Generative AI (GenAI)

1. Course Overview
2. What is Generative AI
3. GenAI VS Predictive AI
4. GenAI Use Cases
5. AI Readiness Index
6. Case Studies of GenAI Deployment
7. Ethics of GenAI

Course Schedule

Part 1 – Responsible Deployment of Generative AI (GAI) in an Organisation

1. Decision - Is GAI the right solution? Alignment with business objectives. Accessing the GAI use case. What type of GAI? Challenges in deploying GAI. Contractual considerations.
2. Design & Development - GAI solution considerations? Different approaches, issues and trade-off (e.g. train own LLM vs customised commercial LLM vs outsourced solution. Cost considerations. Responsible AI approaches
3. Deployment - GAI governance issues. Approaches to GAI risk management and enterprise use. Common errors in using GAI in the organisation.
4. Case studies – Examples of effective and problematic GAI use cases (e.g. chatbot, productivity support, etc) in various domains (e.g. education, legal, healthcare, etc)

Learning Outcomes

ILO 1	Relate various technical and operational issues relevant to the design and deployment considerations of a generative AI solution to a given use case.
ILO 2	Describe the governance and risk management issues related to the responsible implementation of GAI systems with an enterprise.
ILO 3	Describe the different types of security and safety challenges posed by GAI systems and be able to propose appropriate mitigation strategies, techniques and processes to manage such security and safety risks, both during and post deployment.

Course Assessment

End Quiz (only part 1)

MCQ, single answer and multiple answers questions

Project Slides and Sharing

This project will involve investigating a case study of deployment of GenAI tools in an organization, evaluation of its results and measures.

Deliverables are a deck of slides and sharing on next sat session (I will contact you to ask you to share)

Course Assessment Breakdown – Part 1

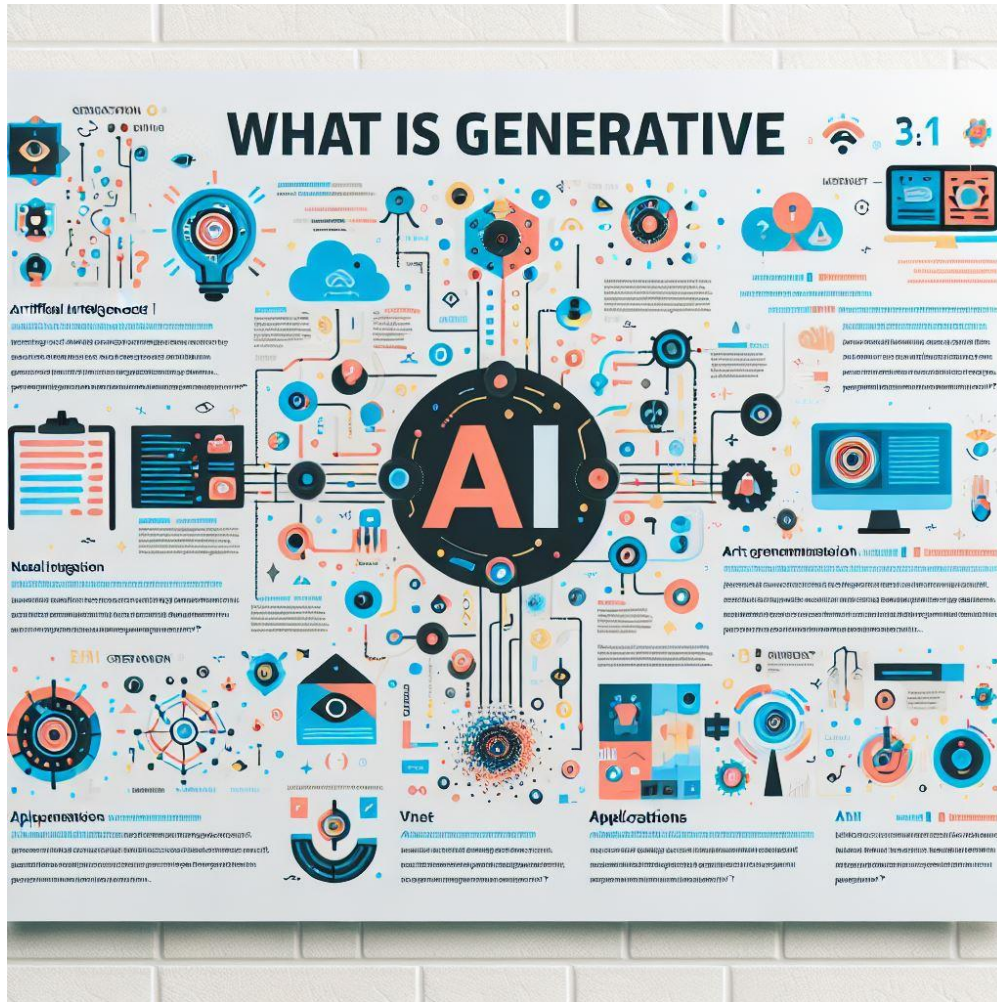
Assessment 1 – Online Quiz 1 (13%)

Assessment 1 is a 30 minutes quiz, to be done on 14 March 2026, Sat

Assessment 2 – Project Assignment (20%)

Details released on NTULearn

What is Generative AI?



Generative AI learns the underlying, finds patterns and structures from data. Then uses it to create new data

Generative AI boomed in the early 2020s due to transformer based deep learning, and we are still amid the boom, with new technologies emerging everyday

LLMs – ChatGPT, Copilot, Gemini

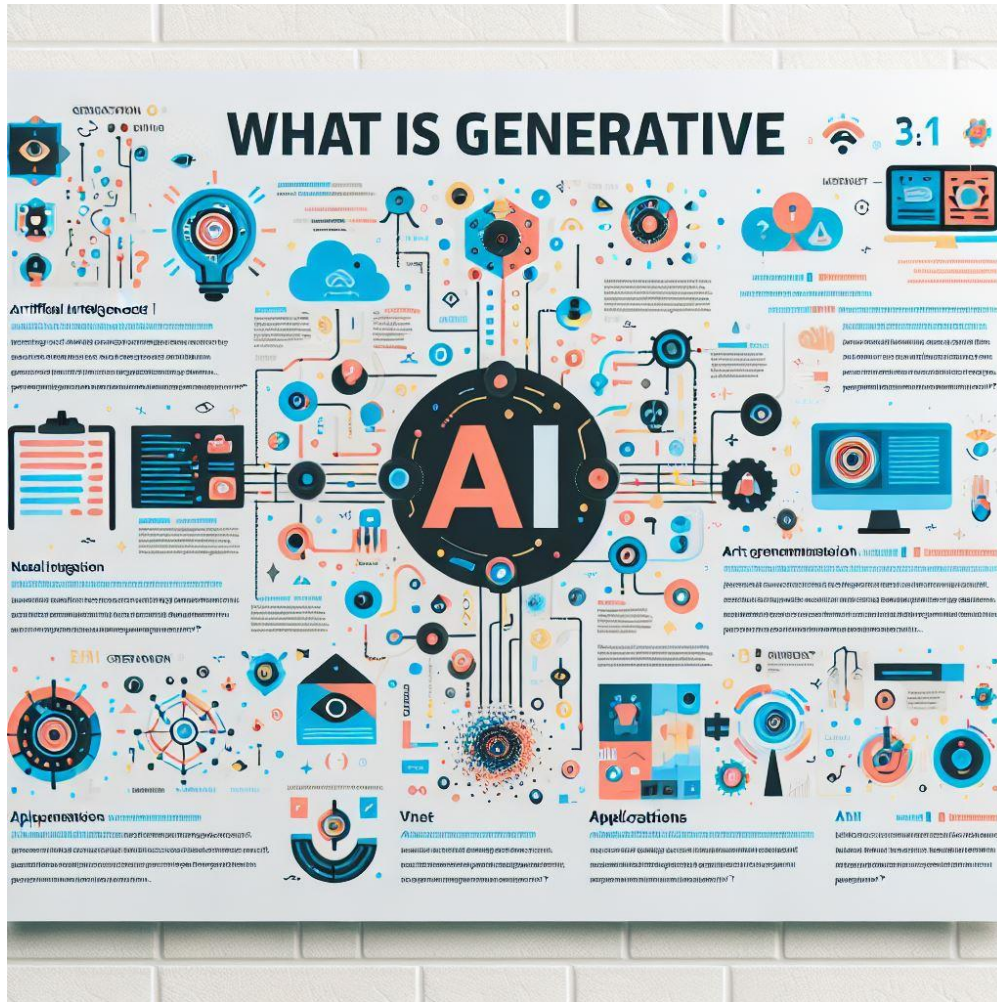
Image – Stable Diffusion, Midjourney, DALL-E

Music – Chuck, Jukedek, MorpheuS

Video – SORA, RunwayML, Make A Video

3D Spatial Models – Computer Aided Design

What is Generative AI?



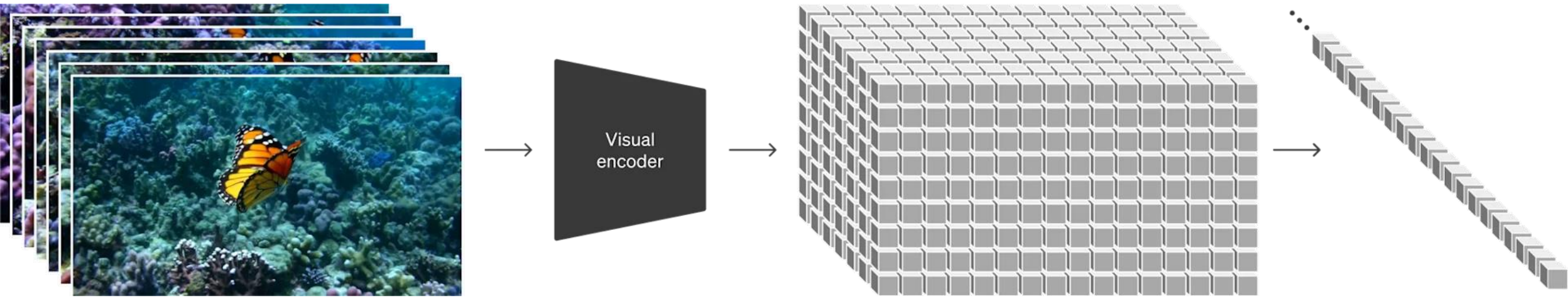
Generative AI can be used to do data augmentation, creating new data in some areas where data is limited

These can also be applied to coding assistants, drug discovery, create 3D environments and personalities

Generative AI also gave rise to AI agents, which can help to automate many tasks that we can do online, including creating events, purchase goods and services etc

AI agents are trending due to advances in LLMs powering them and opening the possibilities of what they can do

Generative AI: Sora



OpenAI SORA is a text to video model that generates short videos based on user prompts.

SORA uses patches to represent visual data, similar to LLMs using text tokens.

At the high level, videos are turned into patches by first compressing them into lower dimensional latent space, then decomposing into spacetime patches

Generative AI: Sora



SORA is a Transformer Diffusion Model.

Diffusion Process: The model is given noisy patches and conditioning information like text prompts, then the model is trained to predict the original “clean” patches.

They are loosely inspired by the Brownian motion in physics where particles collide into each other and each step is a small random walk

Diffusion Models

Denoising diffusion models

- **Forward / noising process**



- **Reverse / denoising process**

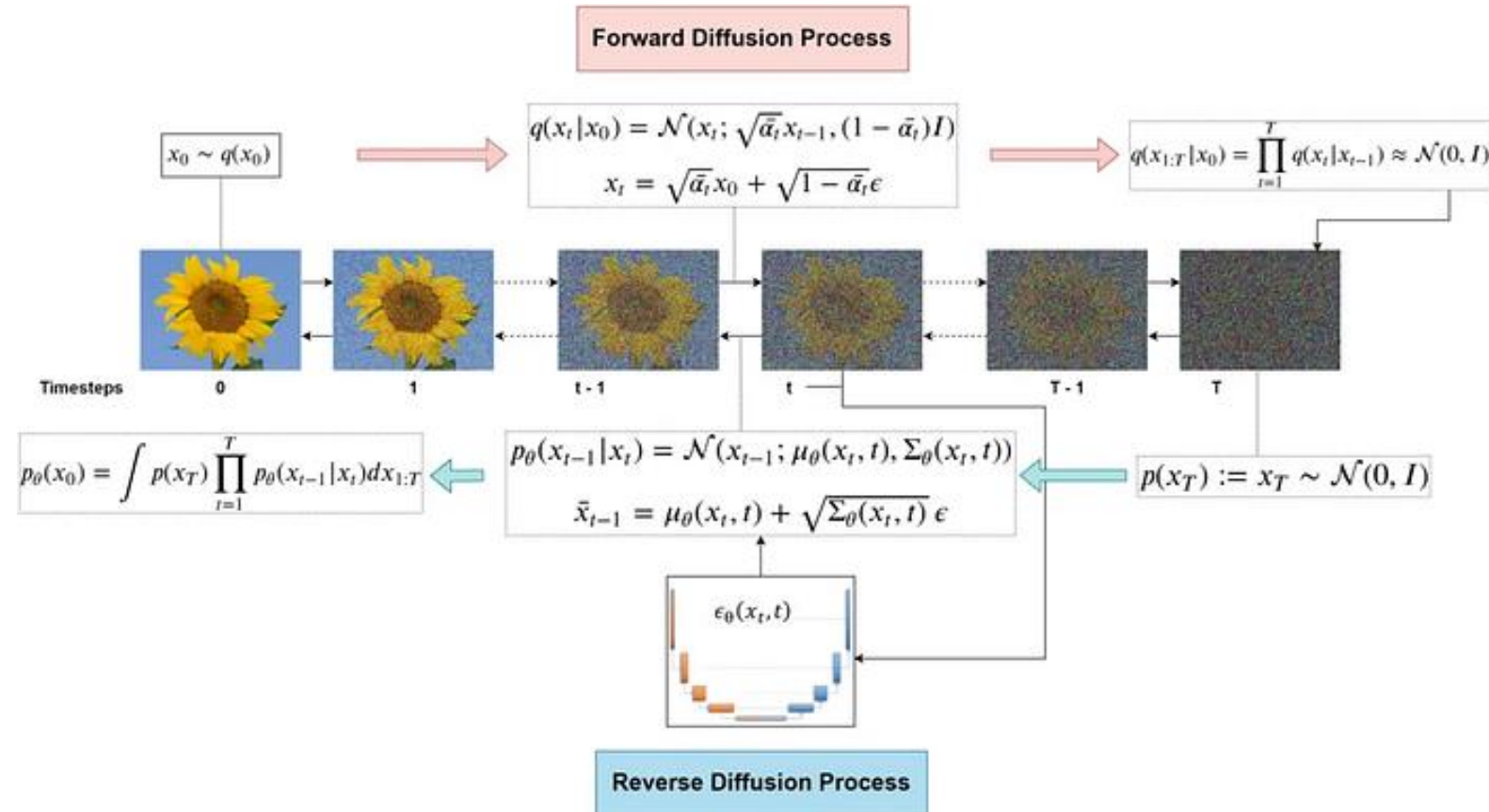
○ Sample noise $p_T(x_T) \rightarrow$ turn into data

Diffusion models start with a dataset and add noise to it incrementally, until it is indistinguishable from actual noise

Reverse diffusion then takes the noisy data and progressively reconstructs the data step by step

After many iterations, the results generated by diffusion models can become high quality, photorealistic pictures

Diffusion Transformers

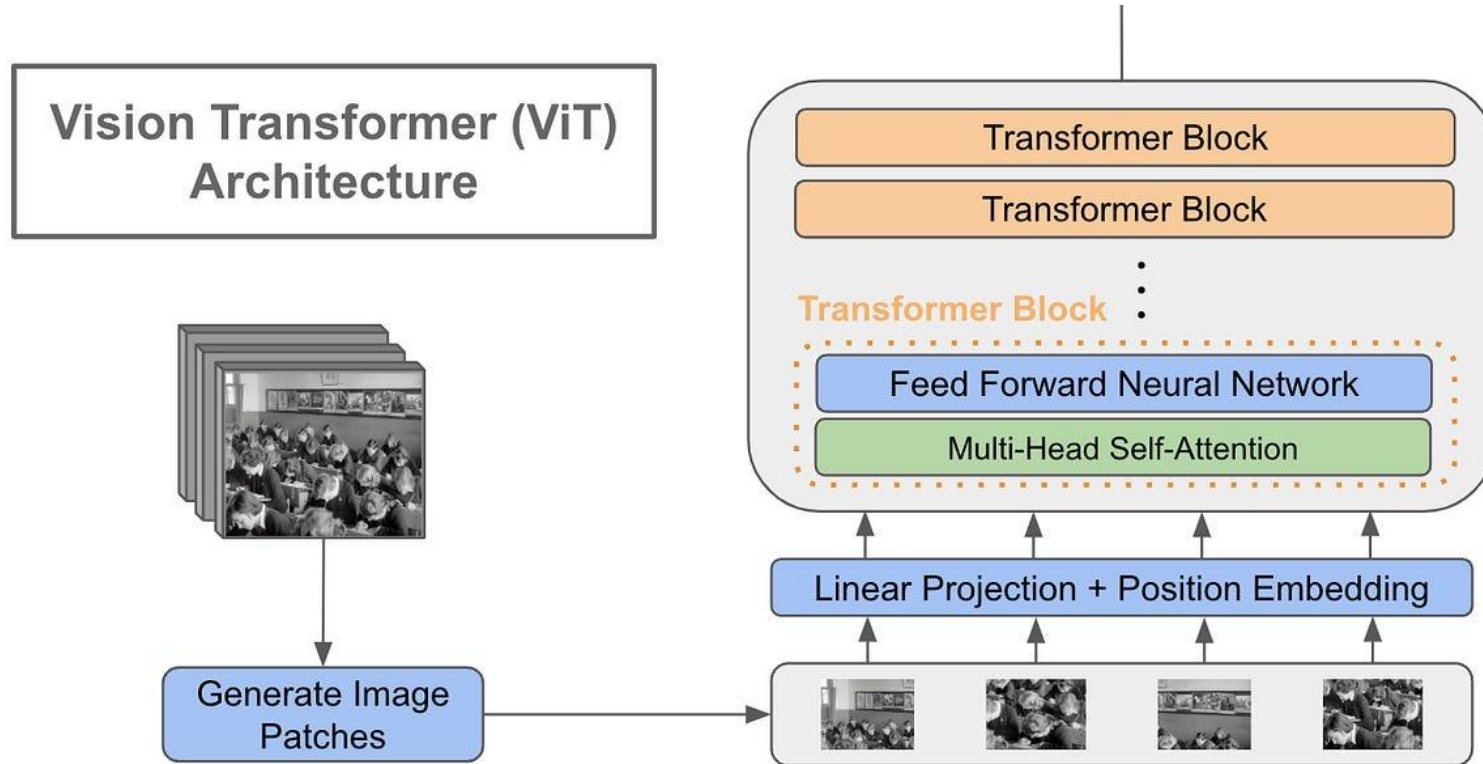


Diffusion Transformers generate data by adding noise gradually to a target image, then reversing that process guided by a transformer.

The transformer guides the process by:

1. Modelling noise, it processes the noisy data as a sequence
2. Self-attention to look at different parts of the input and find signal vs noise
3. Position and timestamp
4. Trained to predict the noise at every time step

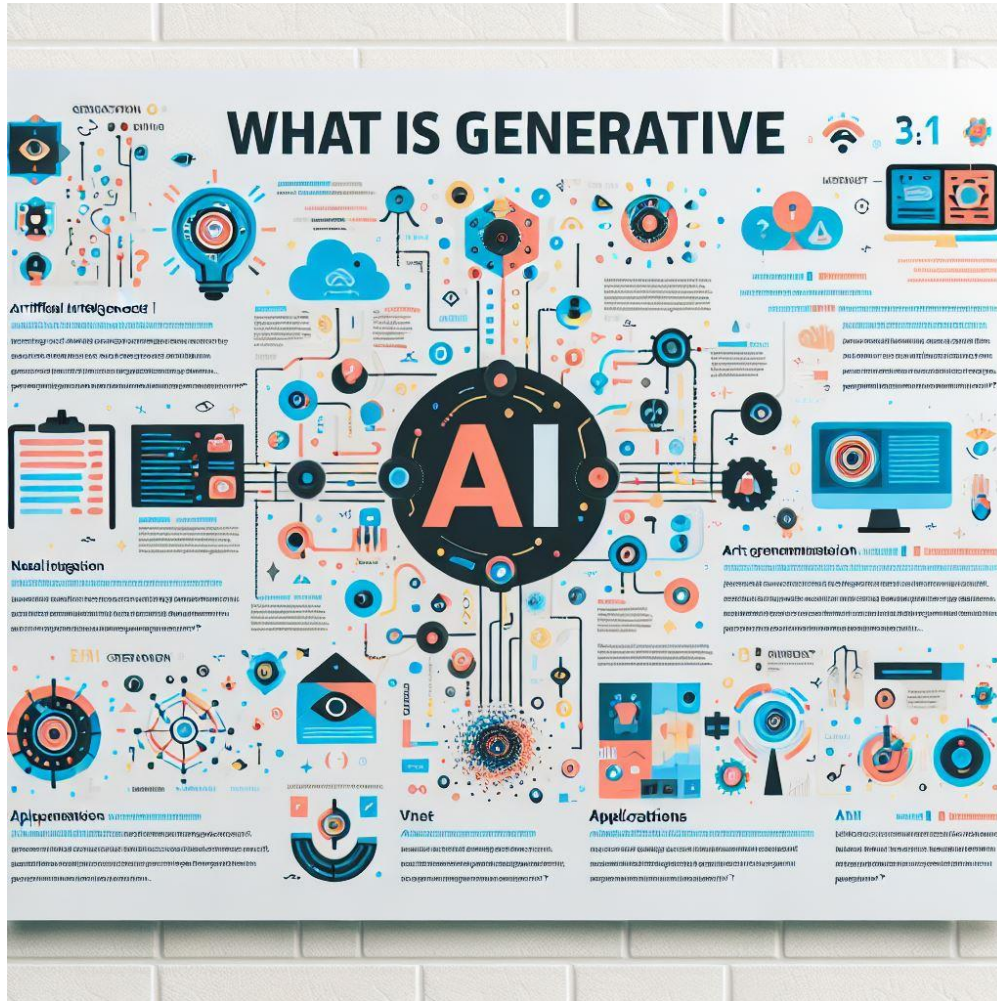
Vision Transformers (Classification)



Vision Transformers work by splitting an image into patches, treats them like tokens instead of using convolutions

1. Image is flattened from a matrix into a vector (like word embeddings in NLP)
2. Positional encodings are added to retain position of each patch
3. A classification token (CLS) is prepended to sequence, output embedding of CLS use for final classification
4. Embeddings are passed through self attention layers to learn relationships
5. Finally CLS output class label

Is Generative AI the Right Tool?



Generative AI is making waves, but is it the right tool for your business?

Problem scoping and use case evaluations

We will be doing the evaluation of the right GAI tool to apply to specific problems, with the help of decision-making frameworks

GAI vs traditional ML/Automation and software engineering

Case studies to learn and reflect from:

Example: GAI in learning and education

GAI in Banking and Finance

Generative AI vs Predictive AI

Generative AI vs. predictive AI

Generative AI creates content and translates data into different formats.
Predictive AI makes predictions and decisions using AI and machine learning techniques.
The two vary in use cases and proficiency with unstructured and structured data.

	Generative AI	Predictive AI
BENEFITS	■ Automates software development ■ Simplifies new content generation ■ Summarizes complex documents ■ Works with unstructured data ■ Creates answers to complex queries ■ Works across text, video, audio, robot instructions and data formats	■ Automates analytics ■ Simplifies complex analysis ■ Streamlines data processing ■ Works with structured data ■ Improves analysis of well-understood use cases ■ Works well for structured and time series data
LIMITATIONS	■ Prone to AI hallucinations ■ Heavy carbon footprint ■ Can be expensive to retrain models ■ Difficult to remove sensitive data from a model ■ Challenging to explain mechanisms underpinning results	■ Bias in underlying data might be amplified ■ Relies heavily on historical data ■ Transparency and explainability can be difficult ■ Overfitting to training data can lead to inaccurate predictions or prediction bias ■ Struggles to distinguish between correlation and causation

SOURCE: ENTERPRISE STRATEGY GROUP

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Generative AI and predictive AI vary in how they handle use cases and unstructured and structured data

Generative AI creates new data, while predictive AI makes predictions

Both have their time and place to be deployed

Explore the benefits and limitations of each, and use that to make decisions on what to use

When to use GenAI?

1. Content Generation at Scale

Use case: Creating text, images, or code with consistent structure

Example: Product descriptions, email replies, code autocompletion

Why GAI: High efficiency and creativity at low marginal cost

2. Unstructured Input → Structured Output

Use case: Summarizing documents, extracting meaning from messy input

Example: Summarizing customer support logs, turning audio into action points

Why GAI: Handles language and patterns better than rules-based systems

3. Tasks Needing Personalization or Variation

Use case: Personalized learning content, dynamic marketing copy

Why GAI: Can tailor outputs using embeddings, persona prompts

4. Rapid Prototyping or Ideation

Use case: Brainstorming designs, writing draft policies, generating test data

Why GAI: Speeds up human creativity cycles

When to not use GenAI?

1. When Accuracy, Reliability, or Legal Precision is Critical

Example: Tax filing, medical advice, legal contracts

Why Not: GAI can hallucinate or be non-deterministic

2. When the Output Requires Factual Integrity

Example: News reports, academic citations

Why Not: GAI may fabricate or cite non-existent sources

3. When You Need Traceability and Auditability

Example: Financial decision systems, safety-critical applications

Why Not: GAI is often a black box with low explainability

4. When the Task is Simple or Rule-Based

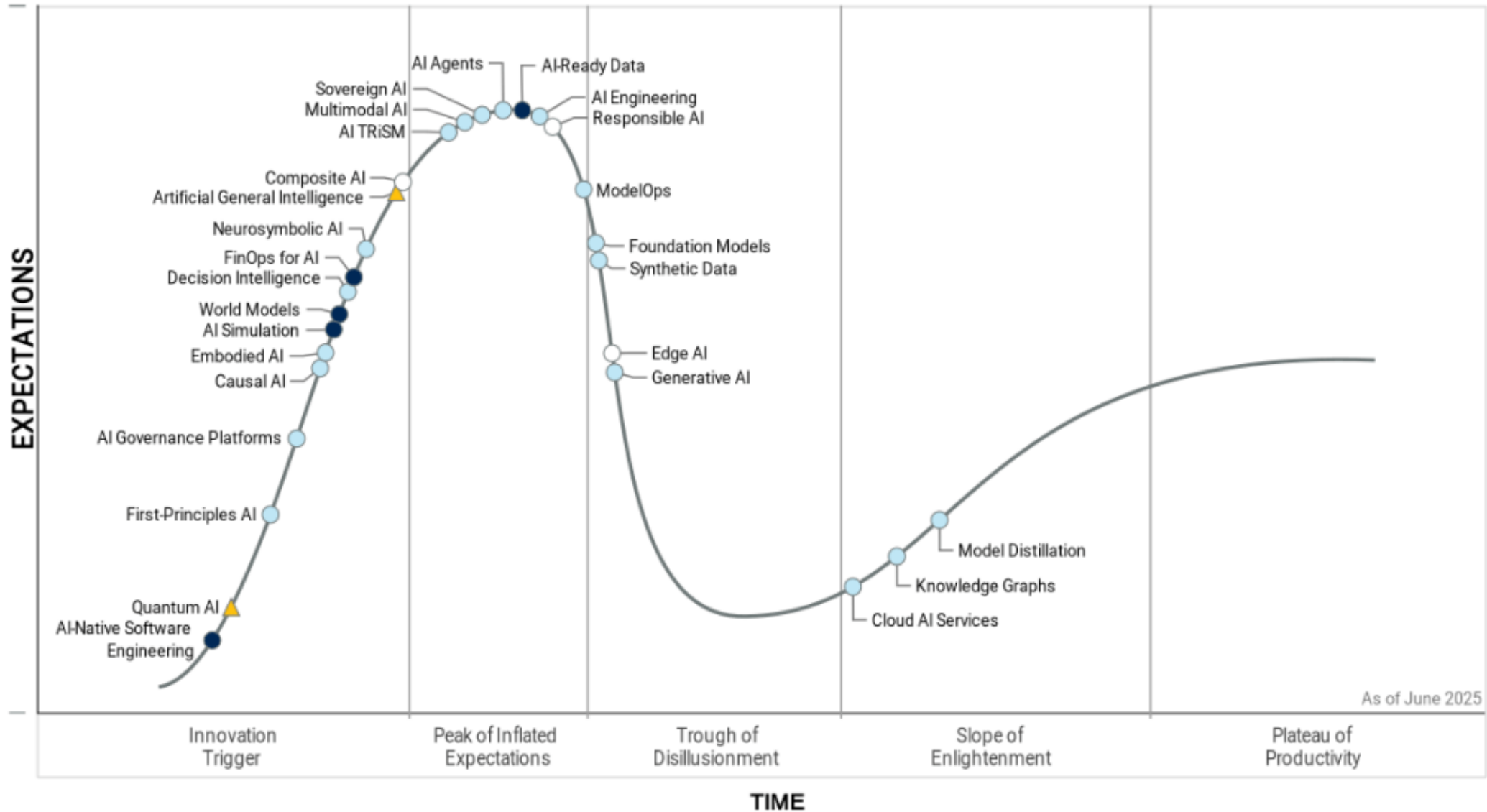
Example: Sorting emails, logging sensor data

Why Not: Simpler automation (regex, workflows) is cheaper, faster, and safer

When to not use GenAI?

Criterion	✓ Use GAI	✗ Don't Use GAI
Unstructured data processing	✓	
High stakes (health, legal, finance)		✗
Output needs creativity or diversity	✓	
Factual correctness is critical		✗
Needs interpretability & traceability		✗
Task is simple and repetitive		✗
You have access to high-quality data	✓	

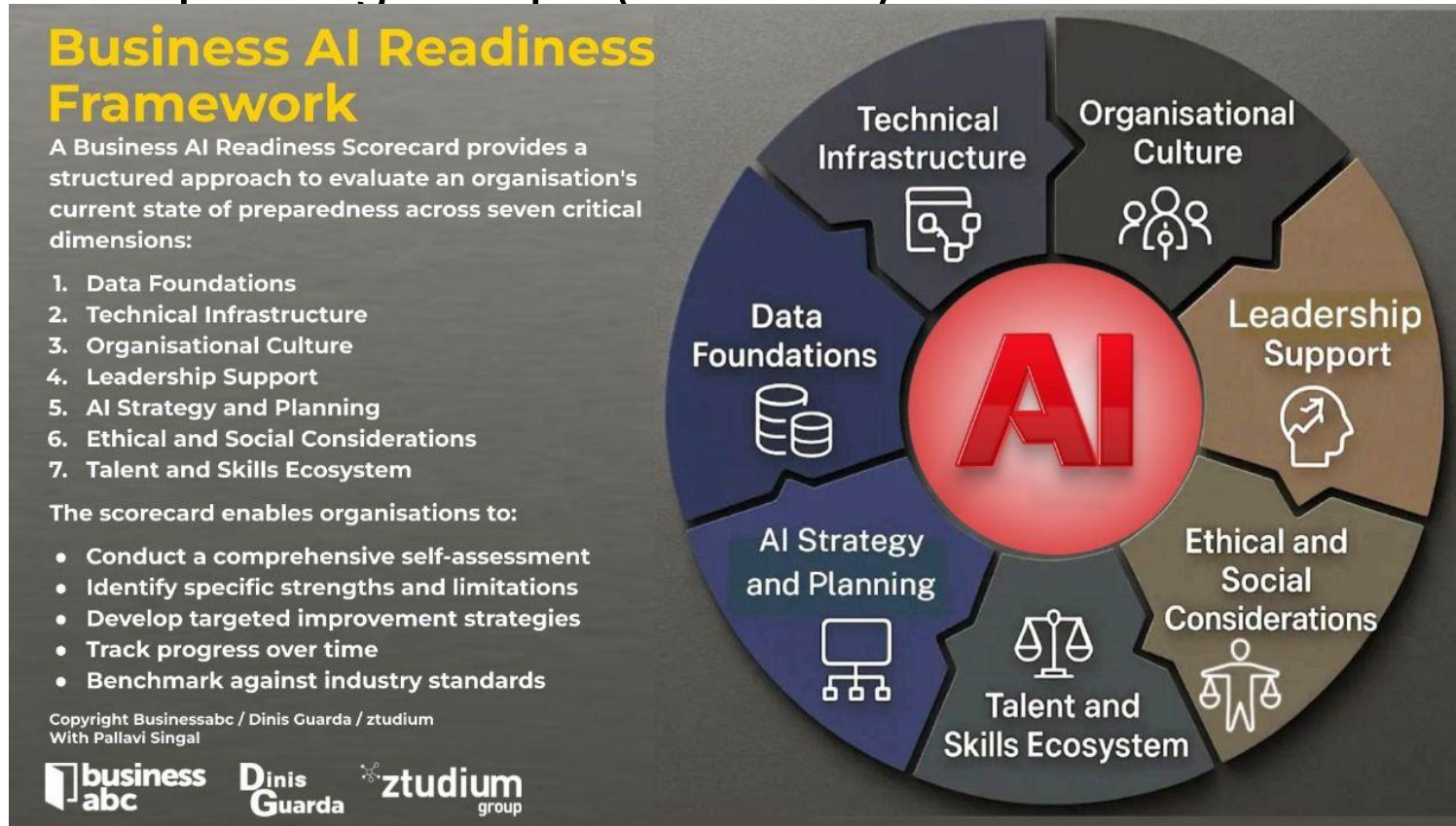
Hype Cycle for Artificial Intelligence, 2025



Plateau will be reached: ○ <2 yrs. ● 2-5 yrs. ● 5-10 yrs. ▲ >10 yrs. ⊗ Obsolete before plateau

Tools to Support Decision Making

1. AI Suitability Scorecards (used in enterprises to evaluate ROI vs risk)
2. Tech Readiness Assessments (to check infrastructure and staff capability)
3. Human-in-the-loop Design Maps (to identify where GAI needs supervision)



Scoring and Interpretation: AISG AI Readiness Index

AI Readiness Index (AIRI) is an industry-focused AI readiness assessment framework developed by AI Singapore (AISG). It crystallises and distils the critical success factors for AI adoption based on hundreds of engagements AISG has with companies across different industries, sizes, and AI readiness.

AIRI allows business units and organisations to assess their AI readiness and identify the gap between their current and desired state. It enables organisations to understand their suitable approaches to adopt AI and implement targeted programmes to increase AI readiness.

Ultimately, AIRI translates abstract concepts into concrete actions to help organisations accelerate their AI adoptions.

AI Singapore uses AIRI to help our companies identify which of our programmes (and our collaborator's programmes) can be leveraged to accelerate their AI journey.

Scoring and Interpretation

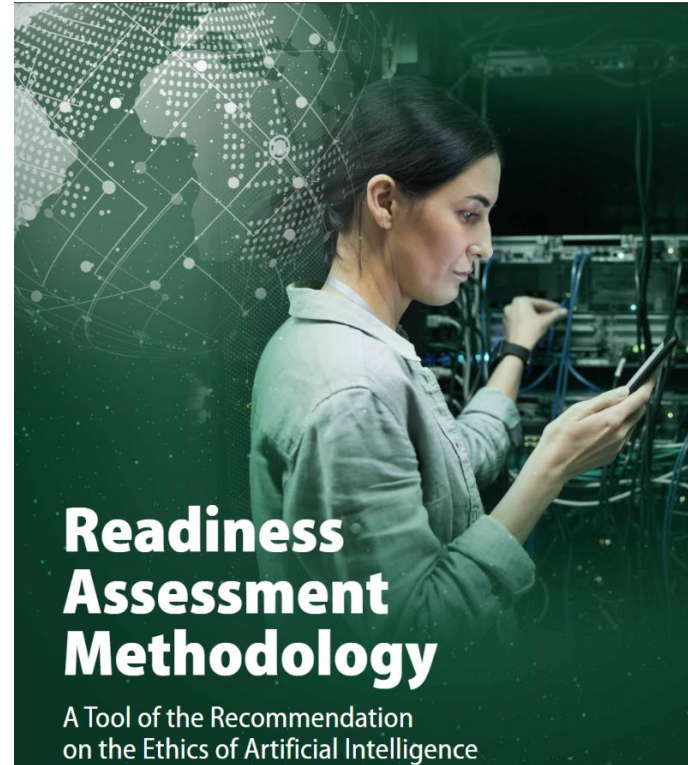
To calculate the readiness score, organisations rate themselves on each criterion, calculate dimension scores by averaging the criteria scores, and then calculate the overall AI readiness score. The overall score will place the organisation in one of the following stages:

- 1.0-2.0: Nascent Stage — Significant work needed
- 2.1-3.0: Emerging Stage — Foundation established but significant development required
- 3.1-4.0: Developing Stage — Moderate capabilities with targeted improvements needed
- 4.1-4.5: Advanced Stage — Strong capabilities with refinement opportunities
- 4.6-5.0: Leading Stage — Exceptional capabilities positioned for AI leadership

Scoring and Interpretation: AISG AI Readiness Index

	AI Unaware	AI Aware	AI Ready	AI Competent
Average Score	Less than 2.5	2.5 to 3.4	3.5 to 4.5	Greater than 4.5
Interpretation	Organisation might hear about AI, but is unaware of AI applications.	Organisation is aware of AI applications and could identify potential use cases.	Organisation has the capabilities to integrate pre-trained AI models into products or business processes.	Organisation has the capabilities to develop customised AI models and solutions for specific business needs.
Characteristics	Wait for vendors to convince with use cases and business value of AI Consume ready-made AI solutions	Actively seek AI solutions to address business needs. Able to identified potential use cases for AI applications, and consume ready-made AI solutions.	Evaluate and seek AI APIs, SDKs and pre-trained AI models for use within business.	Have strategy and roadmap of AI deployment for organisation.
Recommendation	Increase AI literacy of organisation.	Consume ready-made, end-to-end, COTS AI solutions.	Prepare organisation to adopt and integrate AI solution. Broaden understanding of AI to whole of organisation.	Deepen organisational AI capabilities. Broaden understanding of AI to whole of organisation.

UNESCO AI Readiness Assessment Methodology



UNESCO's AI Readiness Assessment Methodology was employed by a ministry of education to evaluate its readiness for AI integration in education.

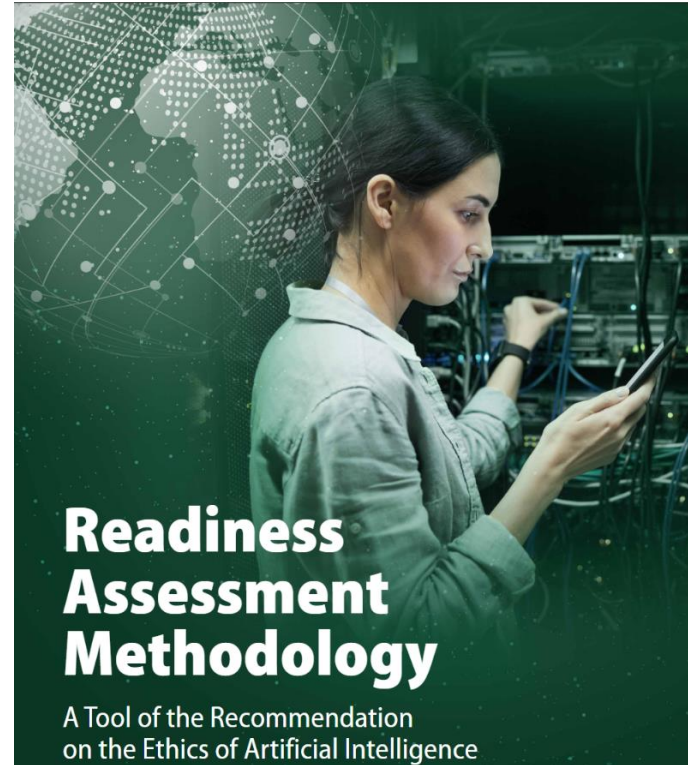
The assessment revealed several key gaps: a moderate policy framework (3.2), underdeveloped infrastructure in rural areas (2.4), strong capacity development through teacher training (3.8), but gaps in ethical frameworks (2.1) and equity (2.3).

UNESCO AI Readiness Assessment Methodology

1. General Questions

- 1.1 Does your government currently have plans to implement the UNESCO Recommendation on the Ethics of AI, through establishing a national AI Ethics Commission or through other measures?
- 1.2 Has your government done any prioritization regarding the specific sectors that would benefit from government intervention (such as regulation, strategies or guidelines, etc) regarding AI?
- 1.3 Does your government inform the public when they are subjected to the use of AI systems that profile or make decisions about them in the provision of public services?
- 1.4 Which ministry/ies are responsible for AI governance?
Please elaborate on the composition of the teams.
- 1.5 Overall, what are the most important challenges to developing AI regulations and policies in the country?

UNESCO AI Readiness Assessment Methodology



UNESCO's AI Readiness Assessment Methodology suggests that AI adoption should always be evaluated against principles such as:

Human rights, Inclusivity, Sustainability, Accountability, Transparency

Case Study 1: DBS Bank Digital Transformation

From Banking Tradition to AI Leadership

How DBS Bank Integrated
Strategy, Implementation, and
Governance for Sustained
Transformation



DBS Bank: AI Readiness & Digital Transformation

From traditional bank > World's Best Bank

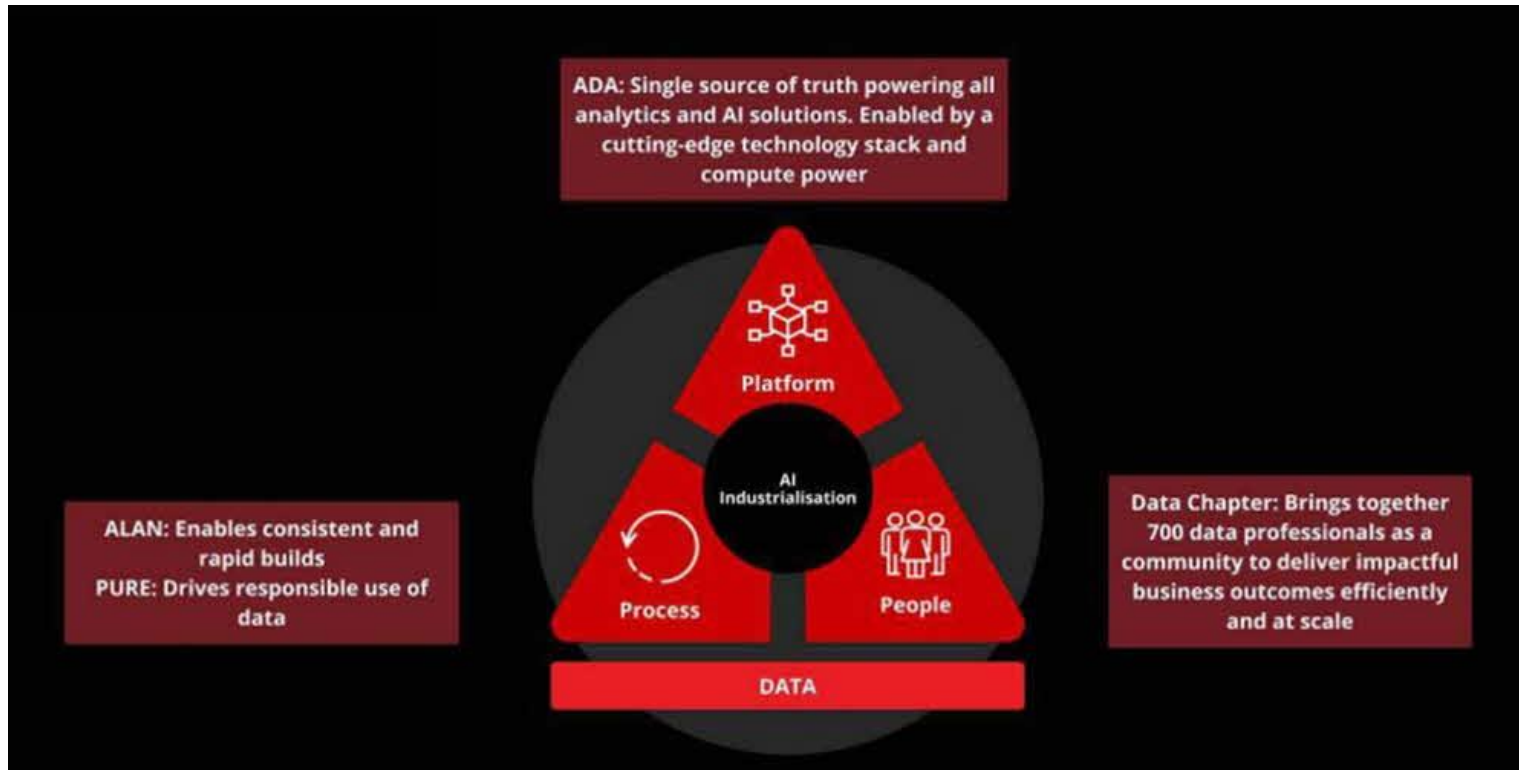
'Live more, Bank less'

Customer-first digital philosophy

AI driving:

- Experience
- Risk management
- Efficiency

Case Study 1: DBS Bank Digital Transformation



Data readiness – unified platforms, data lakes

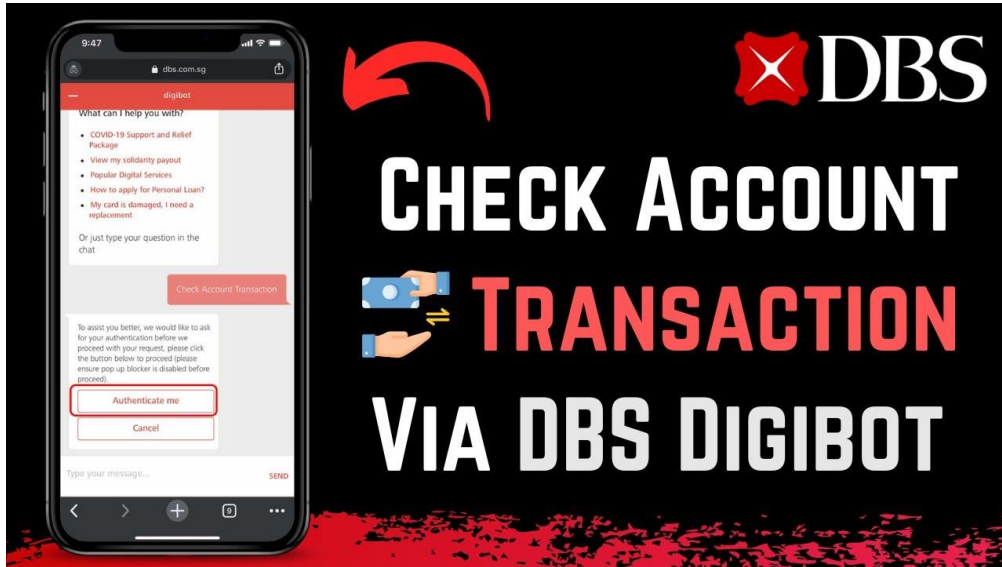
Tech readiness – cloud-native, APIs, ML platforms

Talent readiness – in-house data science, academia links

Governance readiness – ethics, AI council

Change readiness – cultural shift

Case Study 1: DBS Bank Digital Transformation



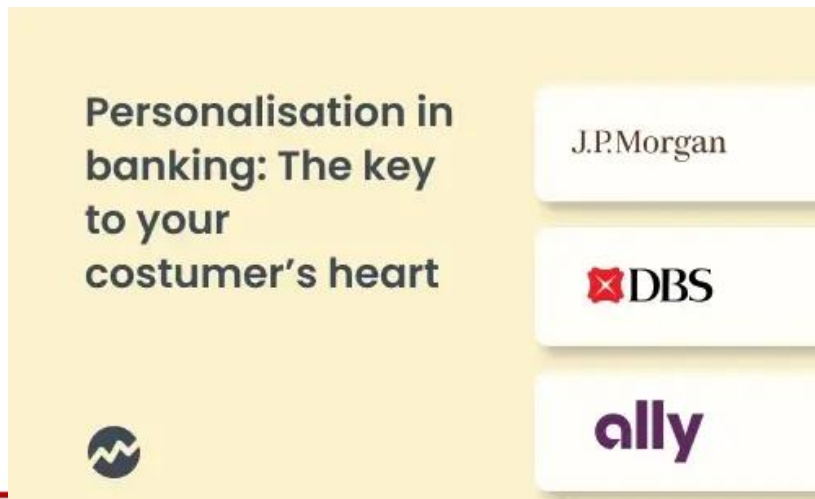
AI powered chatbots like Digibot handle large volumes of customer queries automatically, reducing load on CSO

Hyper-personalisation delivers personalised financial advice based on customer behaviour

AI driven segmentation and Market campaigns

Fraud detection & Anti money laundering with ML

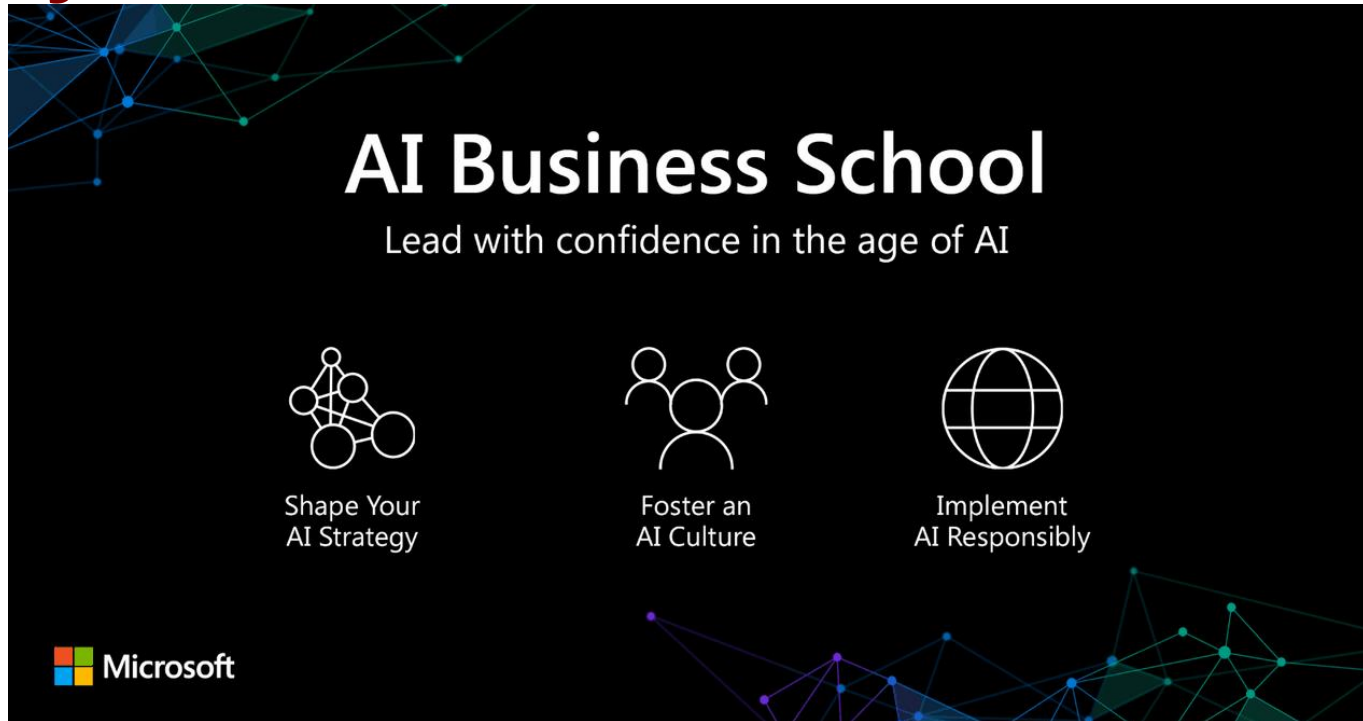
AI driven credit scoring



DBS Pure Framework for Responsible AI



Case Study 2: Microsoft AI Assessment



A multinational retail organisation used Microsoft's Assessment framework to evaluate its AI readiness. The assessment highlighted regional discrepancies in strategy alignment (2.3-4.1), cultural challenges around data-driven decision-making (2.8), and strong capabilities in e-commerce (4.2) but weaker AI adoption in logistics and marketing. The company also recognised the need to strengthen its responsible AI practices (3.0).

Case Study 2: AI Business School Assessment

Step 1: Evaluate Current State

- Assess alignment of AI objectives with business priorities
- Check data access and security measures

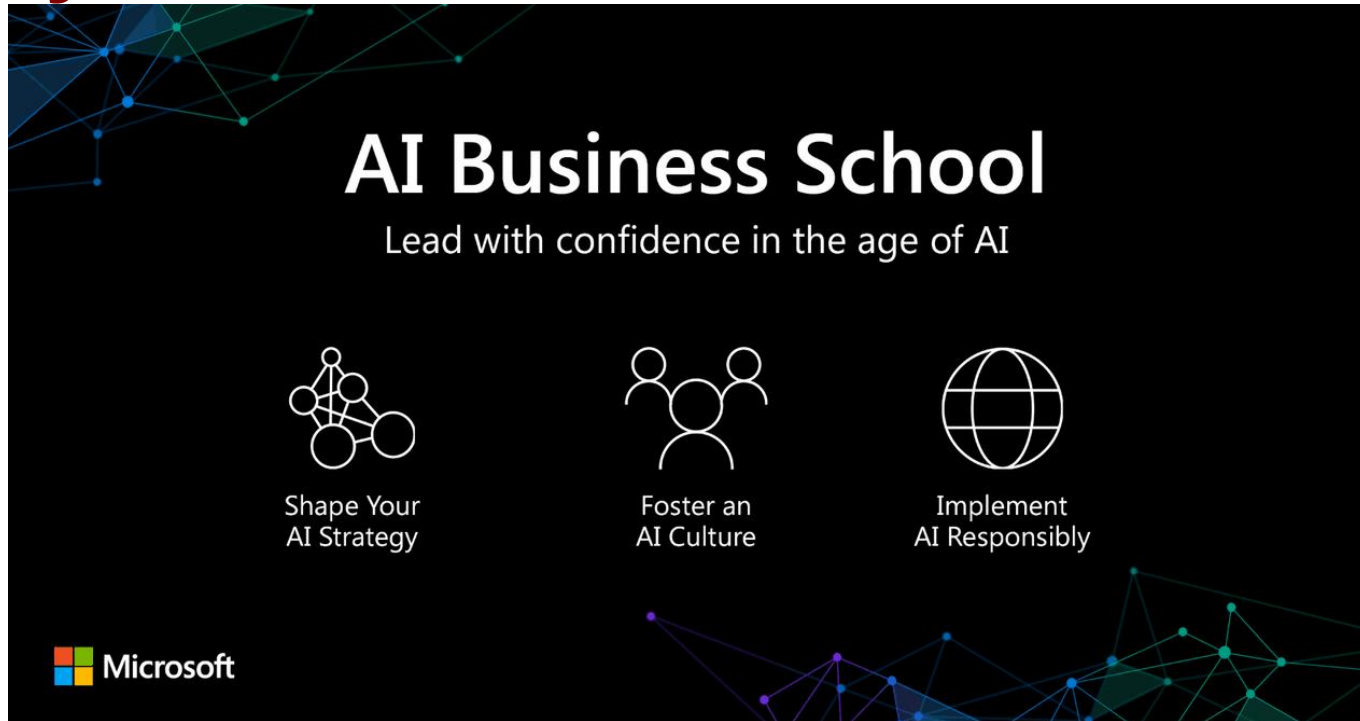
Step 2: Identify Gaps

- Receive scores across key dimensions
- Highlight areas needing focus: strategy, governance, skills

Step 3: Plan Next Steps (5 Stages of AI Readiness)

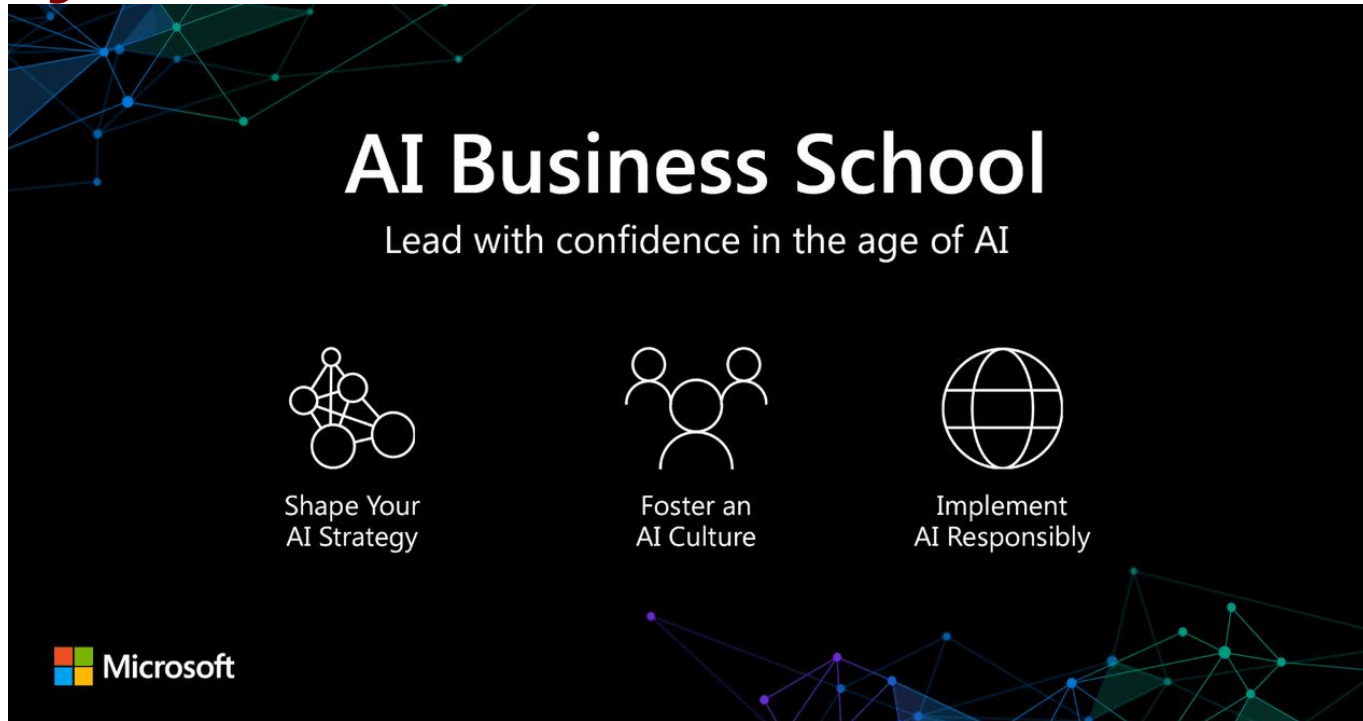
- **Exploring:** Build awareness, learn AI concepts, explore opportunities
- **Planning:** Formalize strategy, prioritize AI projects, learn from peers
- **Implementing:** Secure leadership support, develop skills, execute effectively
- **Scaling:** Expand AI across functions, foster innovation, measure impact
- **Realizing:** Fully embed AI, sustain continuous innovation, create lasting value

Case Study 2: Microsoft AI Assessment



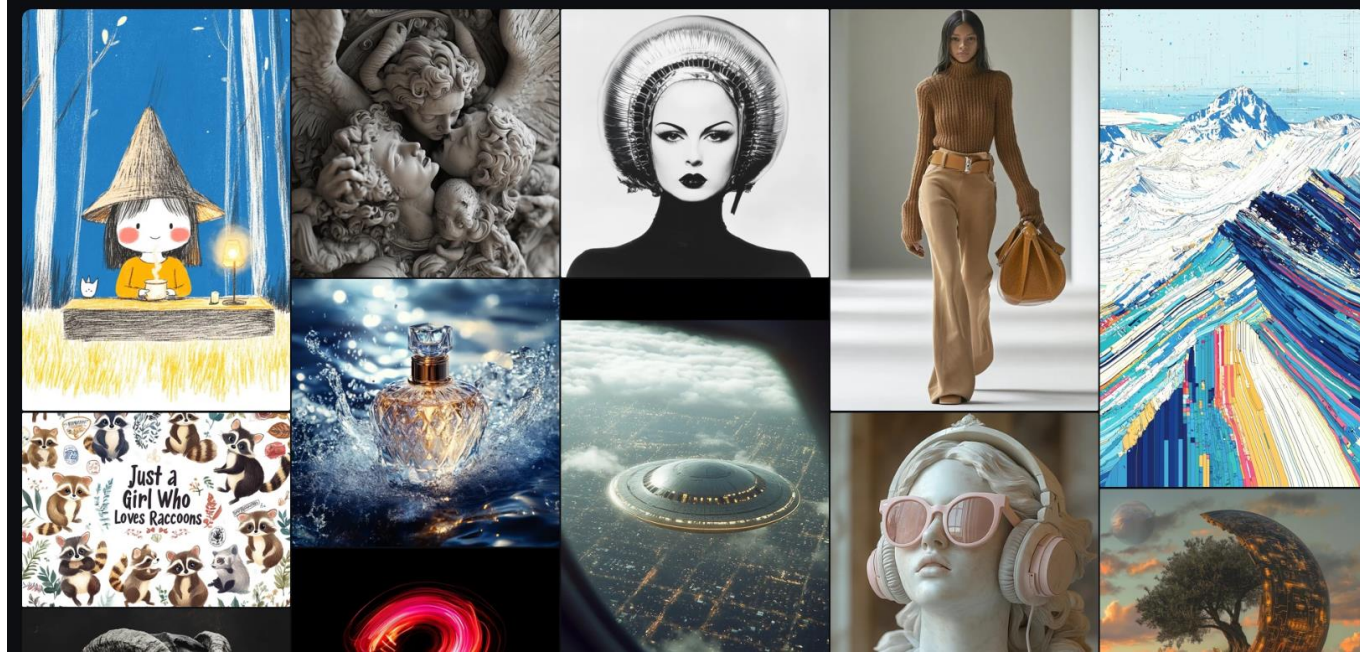
In response, the organisation developed a unified AI strategy that aligned with business objectives and launched a cultural transformation programme led by executive sponsors. A centre of excellence (CoE) was created to transfer AI knowledge across divisions, particularly focusing on underperforming areas. A comprehensive responsible AI governance framework was also established to address ethical concerns and ensure fairness in AI applications.

Case Study 2: Microsoft AI Assessment



After 20 months, the company reported a 40% improvement in AI strategy alignment, a cultural readiness increase to 3.7, and the successful deployment of 15 AI use cases, including inventory optimisation and personalised marketing. These initiatives resulted in £142M in documented business value. This case underscores the importance of a structured transformation programme and responsible AI practices in achieving AI maturity across a large organisation.

Generative AI: What's Next



GenAI can now imitate human creativity, create content and show remarkable capabilities. It can help us to discover new drugs, enhance understanding and predictions about climate change, GenAI can design optimized hardware and software systems, creating a feedback loop for better AI ... However, we need Ethics, Safety and Governance systems to regulate AI

Generative AI: Ethical Considerations

Bias Mitigation

Generative AI models are trained on large datasets that may contain biases, leading to outputs that could perpetuate stereotypes or discriminate against certain groups. Efforts will focus on reducing biases in AI outputs and ensuring fairness across diverse demographics.

Regulation and Safety of Use

Generative AI can be misused for harmful purposes like creating malicious software, phishing emails, or harmful propaganda.

Regulation is key to minimize Governments and organizations are likely to develop and enforce standards to ensure responsible use.

Privacy and Data Security

Generative AI systems often rely on user data to improve personalization, raising concerns about data privacy and security.

Measures such as removing personally identifiable info, secure storage and encryption is needed for privacy

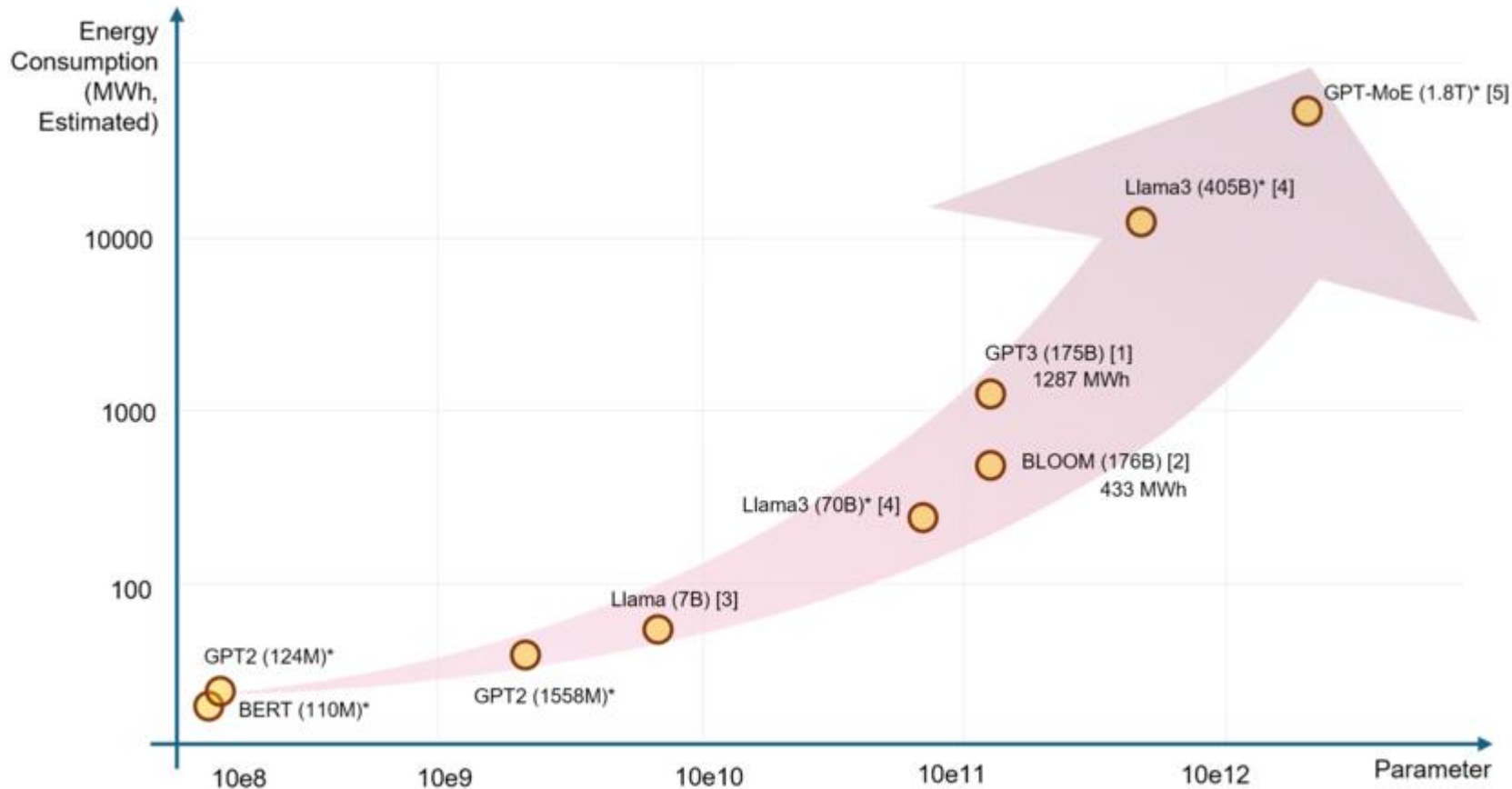
Energy Footprint of LLMs

Model Size (Parameters)	Computational Resources	Training Duration (Hours)	Infrastructure	Training Energy (MWh)	Evaluation Energy (MWh)
7B	8 GPUs (NVIDIA V100)	336	Cloud (Efficient)	50	5
40B	64 GPUs (NVIDIA V100)	672	Cloud (Efficient)	200	10
100B+	1024 GPUs (NVIDIA A100)	1344	Cloud (Standard)	1,287	50
Custom Model	512 GPUs (NVIDIA A100)	1008	On-Premise (Efficient)	800	30

Training these massive LLMs requires extensive computation resources often involve thousands of GPUs running for weeks/months

The smallest 7B parameter LLM requires at least 55 MWh of energy, excluding energy consumption during deployment and user prompts

Energy Footprint of LLMs



A prompt on ChatGPT costs nearly 10 times more than a search on Google.

As the energy consumption of AI grows exponentially, we are looking at alternative sources of energy to power them

One area that big tech is exploring at is nuclear power:

- Consistency and Reliability
- Low carbon footprint
- Scalable

Generative AI: What to focus on?

1. AI-generated medical advice without review
2. Profitable but biased AI system
3. AI's massive carbon footprint

Wrap Up

**The End
Questions?**